Abrash Arshad

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SUMMARY

Aspiring AI and Machine Learning enthusiast with strong Python skills, DSA knowledge, and experience in libraries like NumPy, Pandas, OpenCV. Currently exploring TensorFlow to build intelligent, real-world systems.

EXPERIENCE

AI Engineering Intern

Shahruh Technologies **July 2025 - Present, Lahore, Pakistan**

- Engineered and deployed **3+ AI-powered web applications**, including Transformer architecture explainers and a Retrieval-Augmented Generation (RAG) chatbot, supporting real user interaction with AI systems.
- Leveraged Transformer and attention-mechanism principles (2023 updated *Attention Is All You Need*) to **design and implement scalable AI system workflows** used across **multiple user-facing applications**.
- Implemented **image, audio, and video watermarking modules**, contributing to content protection features across **3 distinct media pipelines**.
- Supported **AI system architecture, frontend integration, and rapid prototyping**, accelerating feature validation and reducing development iteration cycles.

PROJECT

Transformer Learning Hub

Educational Project • learn-transformers.netlify.app/ • November 2025 - January 2026

- **Developed** an interactive, end-to-end visualizer using HTML, CSS, and JavaScript to decompose the entire word generation process of Large Language Models into distinct, interactive stages.
- **Translated** the complex mathematical foundations of Transformer architectures, including multi-head attention and positional encoding, into **Number,** granular visual steps, aiming to reduce the learning curve for complex AI concepts.
- **Structured** the application to present advanced AI concepts across core modules, implementing responsive design for **100% compatibility** across desktop, tablet, and mobile viewports.

Speech Recognition Analysis Tool

University Project • github.com/Abrash-Official/Speech-Recognition-Analysis • April 2025 - May 2025

- Transcribed live audio input to text using SpeechRecognition and PyAudio, enabling real-time speech analysis.
- Measured Word Error Rate (WER) with jiwer to evaluate transcription accuracy and highlight specific errors.
- Processed over 100 audio files, identifying and classifying substitutions, deletions, and insertions in transcriptions.
- Improved recognition performance by up to 20% through ambient noise adjustment and optimized timeout handling.
- Generated a structured HTML report to present transcription results and error insights in a clean, readable format.

Portfolio

Self-Initiated • abrash.dev/ • July 2024 - Present

- Built and deployed a multi-page portfolio using HTML, CSS, and JavaScript, showcasing 6+ academic and technical projects.
- Embedded external links to LinkedIn, GitHub, and email contact, increasing visitor interaction by 40% in the first month.
- Structured content into 4 core sections (About, Skills, Projects, Contact), reducing page exit rate by 30% through improved navigation.
- Implemented responsive design for 100% compatibility across desktop, laptop, and mobile viewports.
- Launched via Netlify, achieving 100% uptime and average loading time under 1.5 seconds for users across 3 regions.
- Regularly updated content to reflect 5+ newly added skills and ongoing project milestones, ensuring relevance and engagement.

EDUCATION

Bachelor of Science in Computer Science - CS

Institute for Art and Culture • Lahore, Punjab, Pakistan • October 2023 - October 2027 • 5th Semester (3.6/4)

Faculty of Science, Pre-Engineering

Kips College • Raiwind, Lahore, Pakistan • 2023

SKILLS

Skills: Programming Language:
Python, C++, Vibe Coder

Tools
Technical Skills: Git, GitHub, Numpy, Pandas, Matplotlib, Leetcode

Soft Skills:
Self-Motivation, Initiative, Curiosity, Lifelong Learning, Problem-Solving, Time Management, Goal-Oriented Mindset, Collaboration-Ready